

JACOB N PICKOS

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Research and Career Interests: *Embedded Systems, Edge AI, Secure Hardware Design, AI-native Operating Systems*

Education

University of Michigan

Ann Arbor, MI

Computer Engineering BSE, 3.65 GPA

August, 2024 - May, 2027

- **Selected Coursework:** Embedded System Design, Digital Integrated Circuits, Digital Logic Design, Quantum Computing, Computer Organization, Electronic Circuits, Web Systems, Signals and Systems, Data Structures and Algorithms, Discrete Mathematics, Principles of Optics, Intro to Nuclear Engineering
- **Awards:** Michigan Engineering Continuing Student Scholarship (twice awarded)

Research Experience

Oak Ridge National Laboratory

Oak Ridge, TN

AI Engineering Research Intern

May 2026 - July 2026

- First author on conference paper submission to IEEE Big Data 2026 – Privacy and Security of Big Data Special Session (**currently under review**).
- Implemented a full-stack retrieval-augmented generation application with a PostgreSQL vector database to dynamically select between local and cloud LLM inference depending on the security classification of retrieved evidence.

University of Tennessee — NSF REU program

Knoxville, TN

Machine Learning Research Intern

May 2024 - August 2024

- Trained a Deep Q-Network from scratch and implemented it into a reinforcement learning agent.
- Developed autonomous vehicle simulations in SUMO to imitate traffic patterns in the presence of emergency vehicles.
- Coded rule-based vehicle movements in MATLAB to control the SUMO simulations.
- Created and presented a research poster titled "Making Way: Using Vehicle-Vehicle Communication and A.I. to Clear a Path for an Ambulance through Heavy Traffic."

Fisk University — NSF CREST (BioSS) program

Nashville, TN

Biochemistry Research Intern

June 2023 - August 2023

- Applied crystallography to further research on superoxide dismutase, a protein found in *S. aureus*.
- Cultured *E. coli* for protein expression, ran nickel columns for protein purification, and tested protein purity using gel electrophoresis. Grew protein crystals via the vapor diffusion method.
- Processed data from .pdb files for use in PyMOL molecular modeling software to visualize protein structures.
- Created scientific presentations and shared findings before a panel of university faculty.

Project Experience

Multidisciplinary Design Program (MDP) — Walbridge

Jan 2026 - Present

- Built a multi-agent verification pipeline between Microsoft Copilot Studio, Power Automate, SharePoint and Excel to judge the quality and grounding of LLM responses after RAG on company data.
- Led an 8-person undergraduate team in an engineering Design Review before a panel of industry professionals.

Michigan Aeronautical Science Association — Apogee Detection Project Lead

Sept 2024 - Sept 2025

- Created `baro_sim`, a Python library with a C++ back end which gives users access to realistic barometers in software, for use in rocket flight simulations.
- Wrote high-quality C++ code for simulating a multi-stage apogee detection system, integrating data from barometric and IMU sensors.
- Operated within a large team-based firmware repository on GitHub, applying best-practice version control.
- Led a Preliminary Design Review for the apogee detection system, inviting contributions from multiple engineers across our team.

Technical Skills

Languages: C++, C, Python, Verilog, MATLAB, ARMv8 Assembly

Tools & Platforms: Git, Unix, Docker, STM32CubeIDE, LTspice, ModelSim, Weaviate, React, Ollama, Flask, Codex